

SKILL – 17

PART 1 — Job Listing & Status Management:

Code:

```
GNU nano 7.2 skill17_jobs.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

#define MAX_JOBS 10

typedef struct {
    int job_id;
    pid_t pid;
    char state[20];
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

void add_job(pid_t pid)
{
    jobs[job_count].job_id = job_count + 1;
    jobs[job_count].pid = pid;
    snprintf(jobs[job_count].state, sizeof(jobs[job_count].state), "Running");
    job_count++;
}

void update_jobs()
{
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-17$ ls
hi skill17_jobs skill17_jobs.c
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-17$ ./skill17_jobs
OS Skill-17: Job Listing and Status Information
Created Job ID 1 with PID 1305
Created Job ID 2 with PID 1306
Created Job ID 3 with PID 1307

Initial Job Listing:
Job ID  PID      Status
-----
1        1305      Running
2        1306      Running
3        1307      Running

Waiting for jobs to complete...

Updated Job Listing:
Job ID  PID      Status
-----
1        1305      Completed
2        1306      Completed
3        1307      Running

Job listing test completed successfully.
```

PART 2 — Foreground Switching / Terminal Control:

Code:

```
GNU nano 7.2 skill17_foreground.c
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;
    pid_t ppid;
    int status;

    printf("OS Skill-17: Foreground Job Control\n");

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        setpgid(0, 0);

        printf("Child process started. PID = %d\n", getpid());
        printf("Child process is running in foreground...\n");
    }

    [ Read 65 lines ]
    ^C Location  ^U Undo  ^M-A Set Mark
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```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-17$ ./skill17_foreground
OS Skill-17: Foreground Job Control
Parent identified target job PID = 1803
Transferring terminal control to foreground job...
Child process started. PID = 1803
Child process is running in foreground...
Child process completed.
Foreground job completed.
Updating job state...
Job state: Completed

[1]+  Stopped                  ./skill17_foreground
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-17$ |

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-17$ ls -l
total 52
-rw-r--r-- 1 bhardwaj bhardwaj      1 Oct  7 14:08 hi
-rwxr-xr-x 1 bhardwaj bhardwaj 16456 Oct  9 15:23 skill17_foreground
-rw-r--r-- 1 bhardwaj bhardwaj 1321 Oct  9 15:23 skill17_foreground.c
-rwxr-xr-x 1 bhardwaj bhardwaj 16456 Oct  8 17:44 skill17_jobs
-rw-r--r-- 1 bhardwaj bhardwaj 1918 Oct  8 17:44 skill17_jobs.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-17$
```